Problem A

A Probabilistic Six Degree of Freedom Model for Space Diving

Summary

A 190 kg space diver released near the edge of the atmosphere can remain in rarefied flow for minutes before entering dense air, where drag, heating, loss of attitude, and parachute opening become coupled hazards. We construct a local six degree of freedom model with altitude dependent gravity, the 1976 U.S. Standard Atmosphere, a drag coefficient that varies with Mach, Reynolds, and Knudsen numbers, quaternion rigid body dynamics, and active reaction control. A three node thermal network represents a ceramic outer layer, flexible insulation, and the pressure garment. A drogue and a three stage reefed main parachute complete the descent. Wind uncertainty, Earth rotation, component failures, and uncertain physiological and material limits are propagated through 1000 Monte Carlo trials at each of nine release heights. A nominal 104 km release reaches Mach 3.34, 4.86 g, and 36.35 kW m-2 while active control limits the attitude error to 4.96 degrees. With control disabled, the model approaches a 90 degree error and 17.68 degrees s-1, indicating a flat spin hazard. The nominal parachute sequence lands at 6.00 m s-1 and limits the peak harness force to 3.41 kN. The deterministic 5 g boundary is near 104 km, but only 86.8 percent of uncertain 104 km trials complete a safe descent. Interpolation gives a 95 percent point estimate boundary of 95.7 km. On the tested grid, 90 km is the highest release height whose 95 percent confidence lower bound remains above 95 percent. We therefore recommend 90 km as the screening altitude until aerodynamic, thermal vacuum, and drop test data justify a higher value.

1 Problem Interpretation

The problem asks for the greatest altitude from which a 190 kg combination of person, pressure suit, and parachute can return to Earth successfully [1]. We interpret successful descent as a sequence rather than a single velocity limit. The diver must remain controllable, survive aerodynamic load and heating, deploy the parachutes without exceeding harness limits, reach the ground below a survivable vertical speed, and retain adequate landing accuracy. A release height is accepted only when these conditions remain likely under uncertainty.

This interpretation separates three answers. The nominal limit is the height at which one deterministic trajectory first reaches a chosen load boundary. The probabilistic limit is the height where the estimated probability of safe descent falls to 95 percent. A confidence qualified limit also requires the lower end of a 95 percent confidence interval to exceed 95 percent. The last definition is the most conservative and is used for the final recommendation.

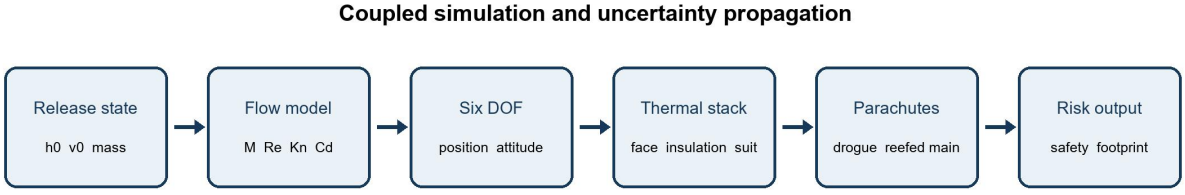


Figure 1 Coupled model from release state to safety and landing outputs

2 Assumptions and Safety Criteria

The model uses the following assumptions. Each one is exposed as a parameter or uncertainty rather than treated as exact.

- The rocket releases the diver with negligible speed relative to the surrounding atmosphere. The initial attitude is broadside to the expected flow, with a small angular rate perturbation.
- Earth is spherical for gravity, while the trajectory is integrated in a local east north up frame. Curvature is negligible over the modeled horizontal range, but Coriolis acceleration is retained.
- Atmospheric density, temperature, and sound speed follow the 1976 U.S. Standard Atmosphere [2]. A multiplicative density uncertainty represents day to day and model variation.
- The pressure suit is rigid enough to maintain shape but flexible enough that its projected area changes with attitude. Three measured human body projections define the area model; drag uncertainty covers suit geometry and unmodeled flow details [3].
- The diver uses a reaction control system above the altitude where aerodynamic control becomes effective. A redundant controller is represented, including a common cause failure probability.
- A drogue stabilizes the low altitude descent. The main parachute has reefed stages and a load limited disreefing rule. A reserve is available, but common cause parachute failures remain possible.
- The landing site is selected using the predicted mean wind drift. Landing dispersion is measured about that biased release point rather than the vertical projection of the target.

2 1 Safety Events

A Monte Carlo trial is safe when its peak aerodynamic and parachute load is below a sampled physiological limit, both thermal nodes remain below their material limits, the harness force remains below 25 kN, touchdown speed is below 8 m s-1, and the required systems operate. Landing within a 10 km target zone is recorded separately. This avoids classifying a survivable off target landing as a fatal descent or hiding a navigation failure inside a physiological statistic.

$$S = S_g \cap S_T \cap S_h \cap S_v \cap S_{\text{system}}, \hat{P}(S) = \frac{1}{N}\sum_{j=1}^{N} I_{S_j} \tag{1}$$

2 2 Principal Quantities

Symbol	Definition	Unit
h	Altitude above mean sea level	m
v and vr	Inertial velocity and velocity relative to air	m s-1
M Re Kn	Mach Reynolds and Knudsen numbers	1
CD and A	Drag coefficient and projected area	1 and m2
q and qc	Dynamic pressure and convective heat flux	Pa and W m-2
T0 T1 T2	Outer layer insulation and inner garment temperatures	K
omega	Body angular velocity	rad s-1
S	Joint safe descent event	Boolean

3 Coupled Descent Model

3 1 Atmosphere and Gravity

Gravity decreases with altitude according to Equation 2. Atmospheric density, temperature, and sound speed are obtained from the standard atmosphere rather than a single exponential fit [2]. This distinction matters near the stratopause and mesopause, where temperature gradients change viscosity, sound speed, and mean free path.

$$g(h) = \frac{GM_E}{(R_E + h)^2} \tag{2}$$

3 2 Aerodynamic Regimes and Attitude Dynamics

The aerodynamic force opposes the relative wind. The coefficient depends on Mach, Reynolds, and Knudsen numbers, while the projected area depends on the direction of the relative wind in the body frame. The three principal projected areas are 1.190, 0.804, and 0.525 m2. These values are consistent with the order of magnitude reported for suited human descent and are dispersed in the uncertainty calculation [3].

$$F_D = -\frac{1}{2}\rho V_r^2 C_D(M, \mathrm{Re}, \mathrm{Kn}) A(\alpha) \hat{v}_r \tag{3}$$

$$M = \frac{V_r}{a}, \mathrm{Re} = \frac{\rho V_r L}{\mu}, \mathrm{Kn} = \frac{\lambda}{L} \tag{4}$$

The continuum drag correlation includes a transonic rise and a smooth supersonic asymptote. When Reynolds number falls below 100000, a viscous correction increases drag. The release phase can have Knudsen number above 0.01, so a sine squared bridge joins the continuum coefficient to a free molecular value of 2.0 across 0.01 less than or equal to Kn less than or equal to 10. This range follows the usual division between continuum, transitional, and free molecular regimes [4].

$$C_D = (1 - w) C_{D,\mathrm{cont}} + w C_{D,\mathrm{fm}}, w = \sin^2\left(\frac{\pi}{2}\mathrm{clip}\left(\frac{\log_{10}\mathrm{Kn} + 2}{3}, 0, 1\right)\right) \tag{5}$$

Translation and rotation are integrated together. The attitude is stored as a unit quaternion to avoid Euler angle singularities. Aerodynamic moments include a restoring term, rotational damping, and a small bias. The reaction control torque is a bounded proportional derivative command that aligns the chest normal with the relative wind. Equation 6 also includes parachute force and the local Coriolis force.

$$m\dot{v} = mg + F_D + F_P + F_C, I\dot{\omega} = M_a + M_{\mathrm{RCS}} - \omega \times I\omega \tag{6}$$

3 3 Pressure Suit Thermal Model

Convective heating is estimated with the Sutton Graves stagnation point relation [5]. A 0.30 m effective radius distributes the broadside heat load. The thermal network has three areal heat capacities: a Nextel class ceramic outer layer, flexible alumina insulation, and the pressure garment. Radiative cooling acts at the outer surface, conduction connects adjacent layers, and an 8 W m-2 K-1 term represents liquid cooling at the inner layer. NASA pressure suit studies identify heating, pressure loss, and uncontrolled rotation as coupled high altitude hazards [6]. Nextel and flexible ceramic insulation provide plausible high temperature material classes, although seams and windows require separate testing [7,8].

$$q_c = k\sqrt{\frac{\rho}{R_n}}V_r^3, C_i\dot{T}_i = \dot{q}_{\mathrm{in},i} - \dot{q}_{\mathrm{out},i} \tag{7}$$

3 4 Parachute Deployment

The drogue becomes eligible below 9 km and Mach 0.8. The main parachute begins below 4 km when downward speed is below 85 m s-1. Its effective area moves through 3.5 percent, 11 percent, and full inflation, matching the scale of demonstrated reefing stages in large parachute systems [9,10]. At every integration step, the allowed increase in drag area is limited by Equation 8. The rule represents a continuously controlled disreefing device rather than an instantaneous opening shock.

$$(C_D A)_{\mathrm{allow}} \le \frac{F_{\mathrm{cap}}}{C_s q}, q = \frac{1}{2}\rho V_r^2 \tag{8}$$

3 5 Wind Drift and System Reliability

The wind model combines an altitude dependent mean profile with a random offset and two smooth vertical modes in each horizontal direction. This construction produces correlated errors along a trajectory. The mean displacement sets the planned release point; the residual displacement defines the landing footprint. Monte Carlo landing dispersion is standard practice for parachute missions [11,12]. Independent backup assumptions can overstate reliability, so the attitude and parachute branches include explicit common cause failures [13]. The component probabilities used here are design assumptions, not measured flight reliabilities.

3 6 Baseline Parameters

Quantity	Baseline	Role
Total mass	190 kg	Specified by problem
Principal projected areas	1.190 0.804 0.525 m2	Attitude dependent drag
Outer layer and inner limits	1370 K and 322 K	Thermal pass criteria
Insulation	30 mm at 0.040 W m-1 K-1	Post pulse heat soak
Main parachute	56 m2 with CD 1.50	Terminal descent
Harness force cap and design load	4 kN and 25 kN	Opening control and strength
Drogue and main initiation	9 km and 4 km	Deployment sequence
Landing zone radius	10 km	Navigation requirement
Monte Carlo samples	1000 per height	Probability estimate

4 Numerical Method and Checks

The six degree of freedom equations are integrated from release to 10 km with an adaptive Runge Kutta method. The parachute phase is then integrated to the ground. A vectorized 0.10 s solver propagates 1000 uncertain trajectories through both phases at each release height. The tested grid is 70, 80, 90, 95, 100, 104, 110, 120, and 140 km. Random seed 2023 makes the reported results reproducible.

Four checks were used. First, the atmosphere reproduces standard sea level density and sound speed. Second, the drag model recovers a low altitude terminal descent of the expected order for a broadside human. Third, the deterministic and Monte Carlo solvers agree on the location of the load transition near 100 km. Fourth, disabling reaction control creates rapid attitude departure while leaving the translational integrator stable. The last check distinguishes a real control hazard from a numerical divergence. These are consistency checks; certification would require wind tunnel, thermal vacuum, and drop test data.

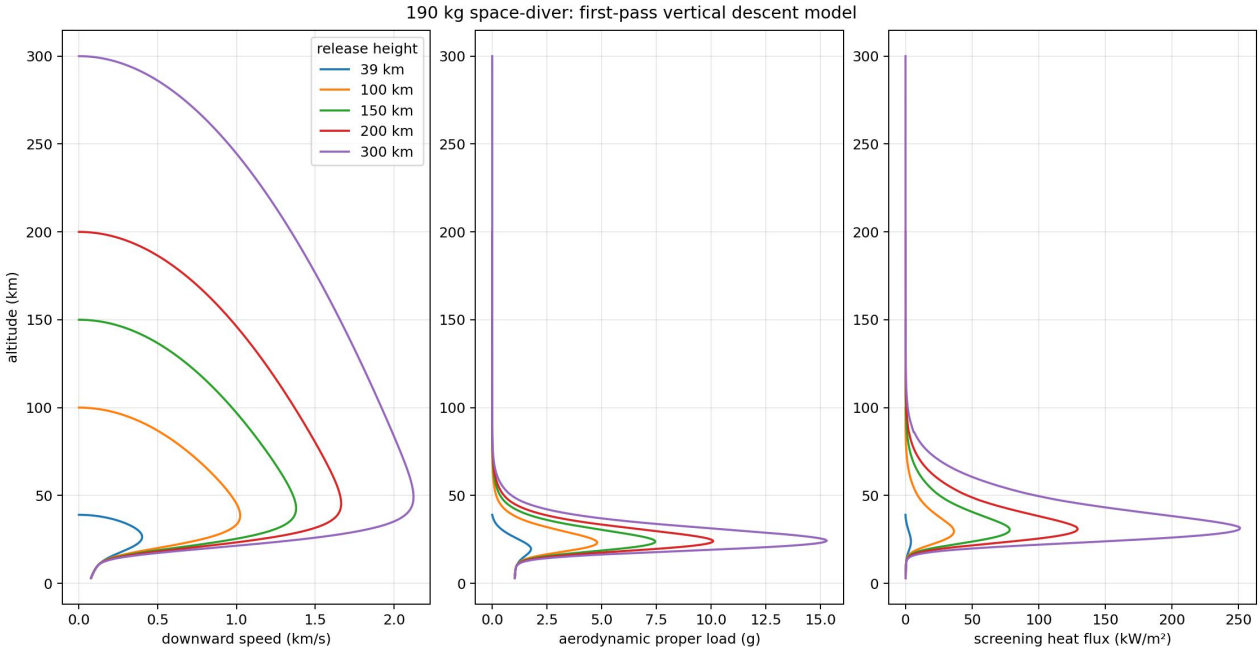


Figure 2 First pass screening profiles used to locate the dangerous altitude range

5 Results

5 1 Nominal Descent from 104 km

The controlled 104 km trajectory reaches 10 km after 190.58 s. Peak speed is 1048.38 m s-1 at Mach 3.345. Maximum Knudsen number is 0.256, confirming that the early descent lies in transitional flow. The peak proper load is 4.865 g and peak heat flux is 36.35 kW m-2. The outer, insulation, and inner temperatures peak at 836.6, 376.0, and 301.7 K. The nominal thermal stack therefore retains margin, but its result depends on idealized layer contact and cooling.

Metric	Controlled	Control disabled
Time to 10 km	190.58 s	190.54 s
Maximum speed	1048.38 m s-1	1048.87 m s-1
Maximum Mach number	3.345	3.346
Maximum proper load	4.865 g	4.868 g
Maximum body rate	2.69 deg s-1	17.68 deg s-1
Maximum attitude error	4.96 deg	90.00 deg
Maximum heat flux	36.35 kW m-2	36.39 kW m-2

5 2 Attitude and Load

Active control holds the meaningful high speed attitude error below 5 degrees. Without control, the error approaches 90 degrees and angular rate reaches 17.68 degrees s-1. The nearly unchanged translational peak load should not be read as safety: the open loop body is tumbling through changing projected areas, and the reduced model does not contain limb loads, visor impacts, or loss of consciousness from sustained rotation. The result makes reaction control a safety critical function rather than an optional performance improvement.

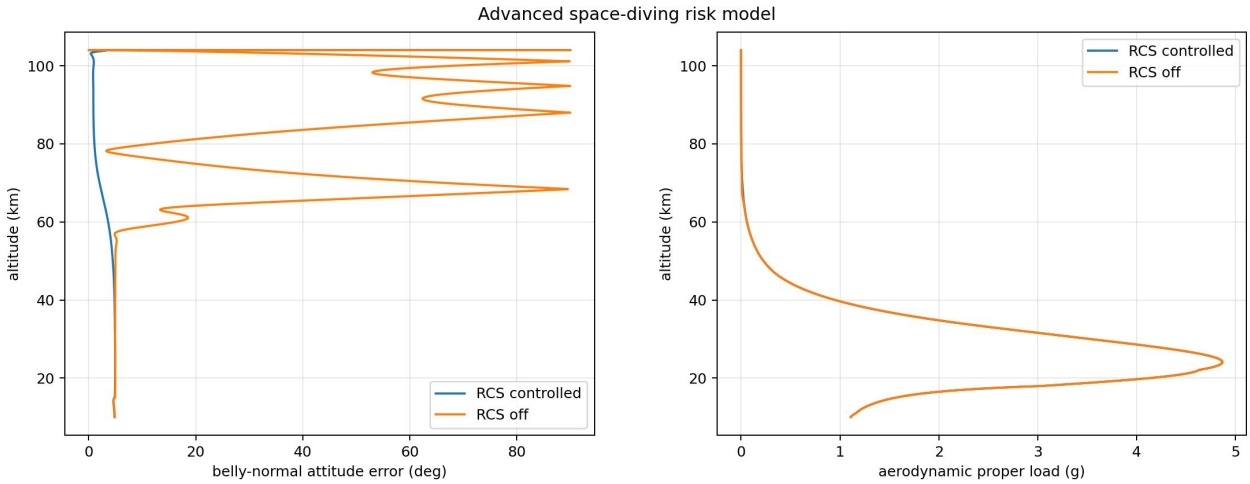


Figure 3 Attitude departure with reaction control disabled and nominal load profile

5 3 Parachute Descent

The nominal parachute sequence lasts 701.68 s below 10 km. The 56 m2 main canopy has an equivalent diameter of 8.44 m and an effective CD A of 84.0 m2. Load limited disreefing keeps the peak harness force at 3.41 kN, or 1.83 g for the 190 kg system. Touchdown speed is 6.00 m s-1. The ratio of the 25 kN screening design load to nominal peak force is 7.32.

5 4 Probabilistic Altitude Limit

The deterministic 5 g boundary lies near 104 km, but uncertainty moves the reliability boundary downward. Table 4 reports the probability of safe descent, its 95 percent Wilson interval, the probability of landing inside the 10 km zone, and the joint probability of both outcomes. Safe descent excludes the landing zone criterion by design.

Height km	Safe	95 percent interval	In zone	Safe and in zone	Load p95 g
70	99.0%	98.2 to 99.5%	93.9%	92.9%	3.46
80	99.1%	98.3 to 99.5%	92.8%	92.0%	3.88
90	98.5%	97.5 to 99.1%	94.8%	93.3%	4.38
95	95.5%	94.0 to 96.6%	93.4%	89.1%	4.63
100	91.8%	89.9 to 93.3%	95.1%	87.1%	4.91
104	86.8%	84.6 to 88.8%	94.7%	82.4%	5.08
110	76.7%	74.0 to 79.2%	93.7%	72.3%	5.45
120	52.7%	49.6 to 55.8%	93.2%	48.9%	5.95
140	14.0%	12.0 to 16.3%	92.2%	13.1%	7.02

The estimated safe descent probability is 95.5 percent at 95 km and 91.8 percent at 100 km. Linear interpolation between these two screening points gives a 95 percent boundary of 95.7 km. The lower confidence bound at 95 km is

only 94.0 percent. At 90 km the estimated probability is 98.5 percent and the lower confidence bound is 97.5 percent. We therefore select 90 km as the confidence qualified recommendation.

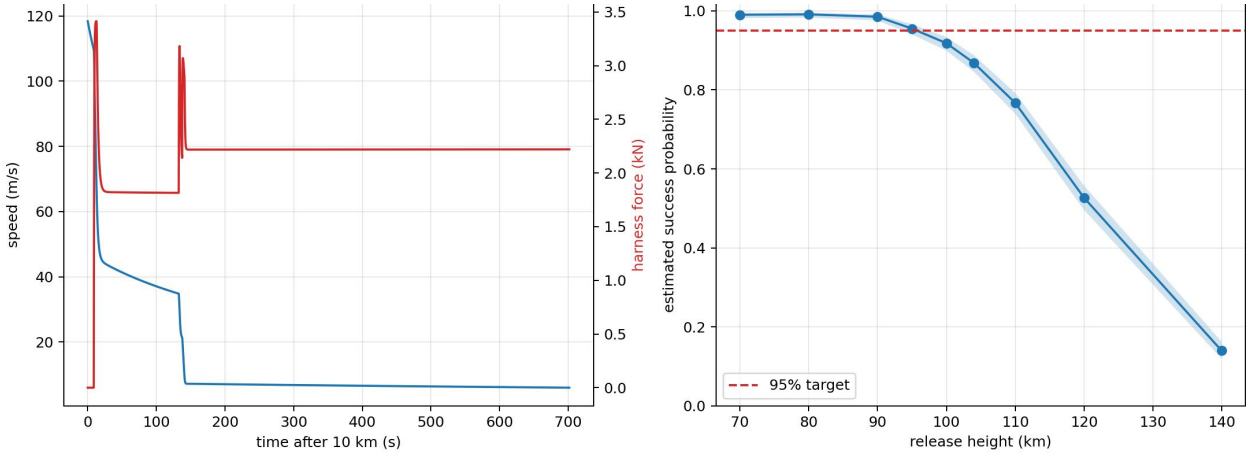


Figure 4 Parachute response and decline of safe descent probability with release height

5 5 Landing Footprint

For the illustrative mid latitude wind profile, the 104 km ensemble drifts approximately 6.30 km east and 1.79 km north before release point compensation. Releasing west and south by these amounts centers the footprint on the target. The remaining 95 percent radius is 10.06 km, and 94.7 percent of samples lie inside the 10 km zone. The 99 percent radius is 13.06 km. A flight plan must replace this illustrative wind field with current radiosonde data and repeat the ensemble before launch.

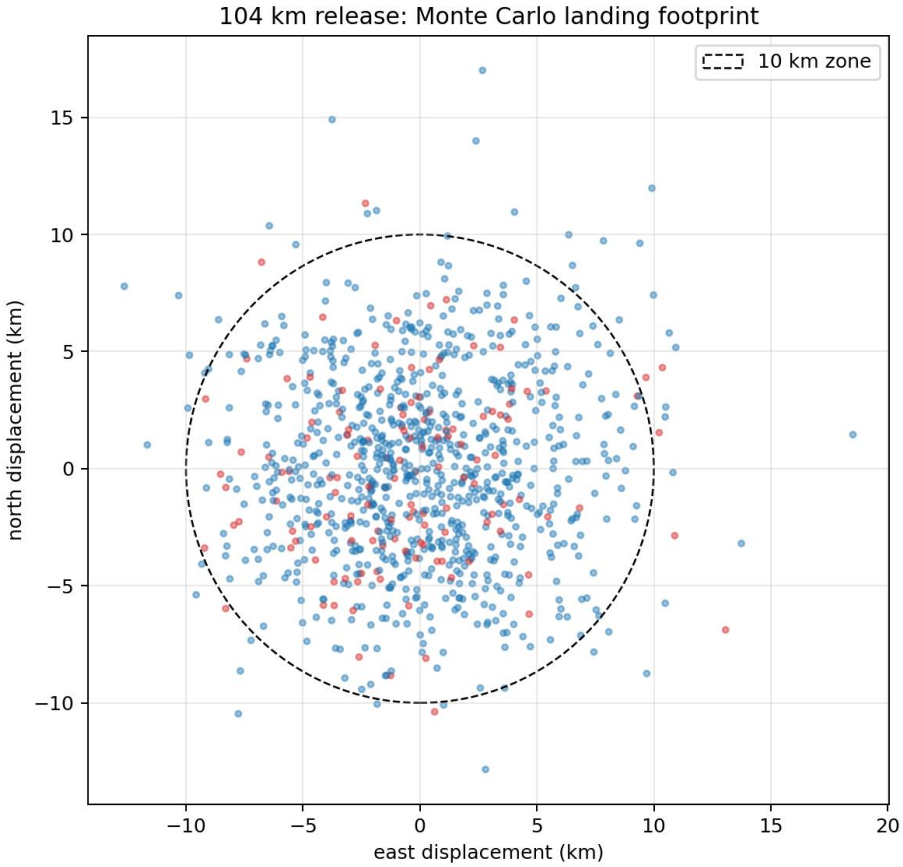


Figure 5 Wind corrected Monte Carlo landing footprint for a 104 km release

6 Dominant Failure Modes

At 104 km, load failure occurs in 12.1 percent of trials and controls the reliability result. Thermal failure occurs in 0.3 percent, touchdown speed exceeds the limit in 0.6 percent, and assumed hardware failures occur in 0.4 percent. No sampled trajectory exceeds the 25 kN harness threshold. The categories overlap, so their percentages do not sum to the total failure probability. Above 104 km, the 95th percentile load grows from 5.08 g at 104 km to 5.45 g at 110 km and 7.02 g at 140 km. The sharp decline in success is therefore physical, not merely a fixed component reliability floor.

Failure event at 104 km	Probability
Physiological load limit exceeded	12.1%
Thermal boundary exceeded	0.3%
Harness design load exceeded	0.0%
Touchdown speed exceeded	0.6%
Outside 10 km landing zone	5.3%
Assumed hardware failure	0.4%

7 Strengths and Limitations

7 1 Strengths

The model keeps the important couplings. Attitude changes area and drag; drag changes load and heating; thermal limits continue to evolve after the heat pulse; parachute deployment changes load and landing speed; wind affects the target but not the definition of physiological safety. The flow model also recognizes that a 100 km release begins outside the continuum regime. Finally, the probability calculation exposes the difference between a nominal boundary and a repeatable safe operating height.

7 2 Limitations

Several inputs cannot be identified from the problem statement. The aerodynamic moment coefficients are preliminary, the high altitude drag bridge is not fitted to this exact suit geometry, and the wind profile is illustrative. The thermal model omits localized conduction through seams, visor absorption, joint geometry, and pressure dependent insulation performance. The human load limit is represented as one sampled scalar even though tolerance depends on direction, duration, restraint, and medical condition. The parachute model does not resolve canopy fluid structure interaction or line elasticity. Component failure probabilities are assumptions rather than fleet statistics. These limitations make the result appropriate for design screening, not human flight certification.

The most valuable next evidence would be a six component force and moment database across Mach and Knudsen numbers, thermal vacuum tests of complete suit joints, and instrumented staged drop tests. Site specific radiosonde ensembles would reduce landing uncertainty. These measurements should update the probability distributions before the recommended height is increased.

8 Conclusions

A successful space dive is limited first by attitude control and aerodynamic deceleration, then by thermal protection, staged parachute deployment, and landing dispersion. The nominal 104 km trajectory remains below a 5 g deterministic boundary and can be landed with a reefed 56 m2 parachute. This nominal result is not sufficiently reliable: only 86.8 percent of uncertain 104 km trials complete a safe descent. The estimated 95 percent boundary is 95.7 km, but its finite sample confidence interval does not support calling 95 km a 95 percent reliable height. Among the tested heights, 90 km is the highest point whose 95 percent lower confidence bound remains above 95 percent. We therefore report 90 km as the present maximum successful descent altitude, conditional on the stated suit, active control, parachute, wind correction, and reliability assumptions.

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Appendix A Computational Algorithm

The calculation used the following reproducible sequence. All constants, distributions, event definitions, and reported outputs are stated in the main text so the algorithm can be implemented independently.

1. Read density, temperature, and sound speed from the standard atmosphere and evaluate gravity at the current altitude.
2. Compute relative wind, Mach, Reynolds, and Knudsen numbers. Evaluate bridged drag and the attitude dependent projected area.
3. Integrate position, velocity, quaternion, angular velocity, and the three thermal nodes until 10 km altitude.
4. Evaluate drogue and main parachute eligibility. Grow the effective drag area through reefed stages while enforcing the harness force cap.
5. Integrate horizontal motion with altitude dependent wind and Coriolis acceleration until touchdown.
6. Evaluate load, thermal, harness, touchdown, system, and landing zone events for each trial.
7. Repeat 1000 trials for each release height, compute Wilson confidence intervals, and locate the 95 percent probability crossing by local interpolation.

Report on the Use of AI

System used. OpenAI Codex, a generative AI coding and writing system provided by OpenAI, was used to organize an English paper draft from the teams independently retained problem statement, model code, numerical results, and source list.

Exact query. 帮我按照作品提交要求生成论文

Retained instruction supplied in the same session. 确保你生成的东西不是直接抄袭参考依据中的内容，而是你在深刻学习后自己写出来的

Full output. This file is the complete output to the exact query, from the first page through this report; no separate AI generated passage is omitted. Numerical values and figures come from the retained simulation outputs. The team must independently verify every equation, number, citation, and statement before submission. This disclosure follows the contest rule requiring the system, exact query, and complete output [14].